Python Programs

Flowcharts and Output Screenshots

Task 1 - Library Resource Distribution

Task 2 - Campus Maintenance Priority

Task 3 - Academic Progression Report

Task 4 - E-Commerce Order Report

Task 5 - Smart Campus Management (menu program)

Task 6 - Campus Event Financial Planning

Task 7 - Campus Event Financial Report

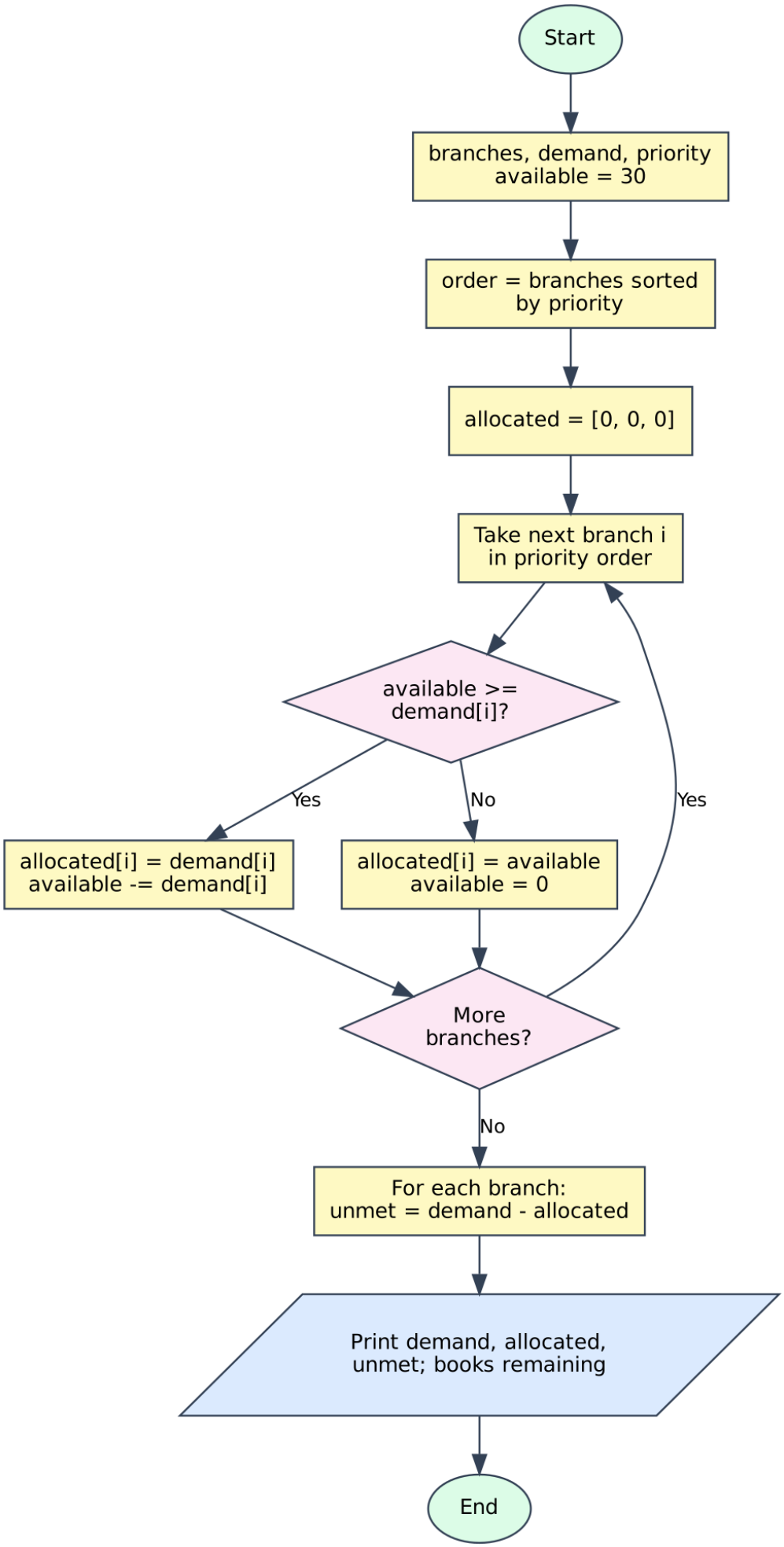
Task 8 - Attendance Recovery Computation

Task 9 - Campus Event Financial Report

Task 10 - Warehouse Inventory Reconciliation

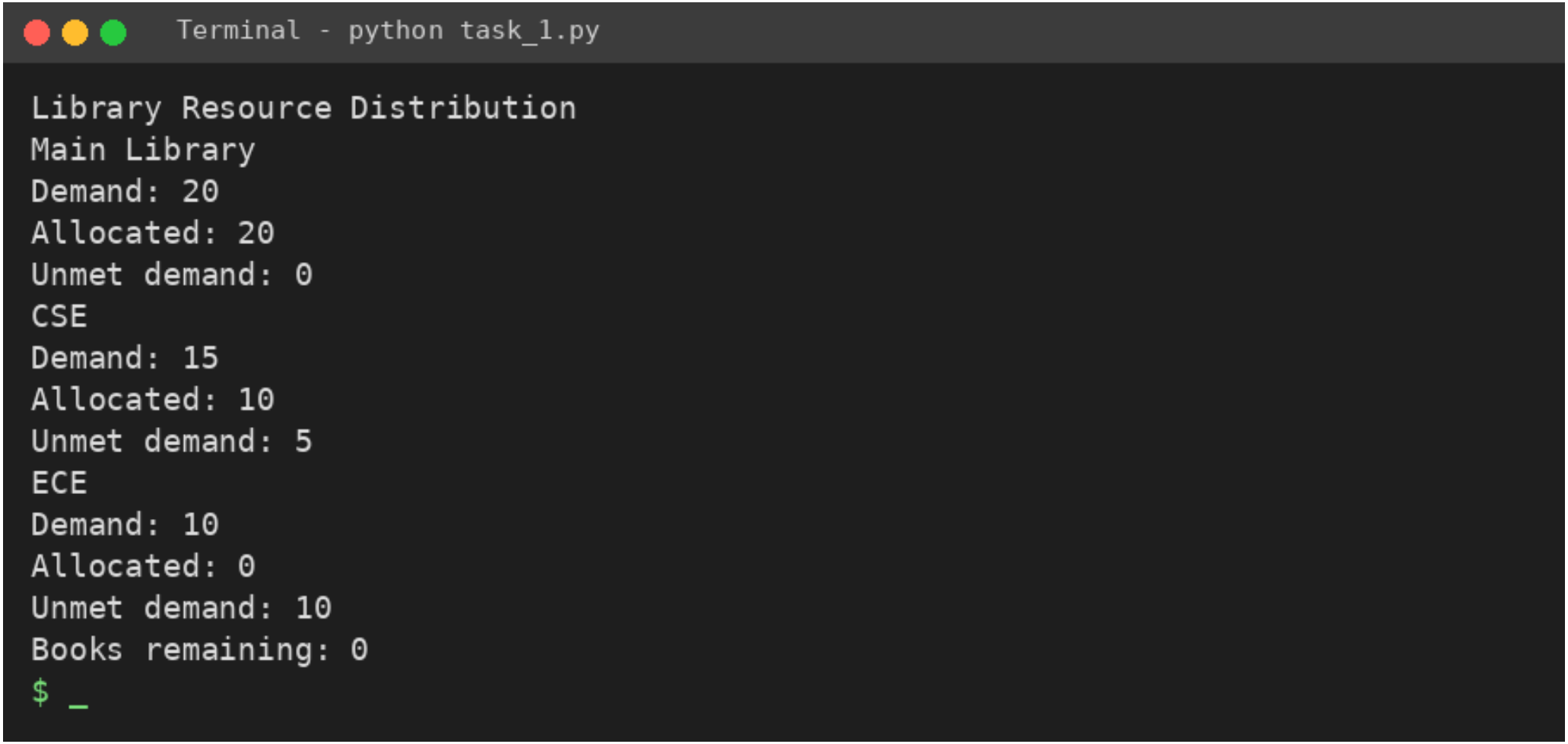
Task 1 - Library Resource Distribution

Flowchart



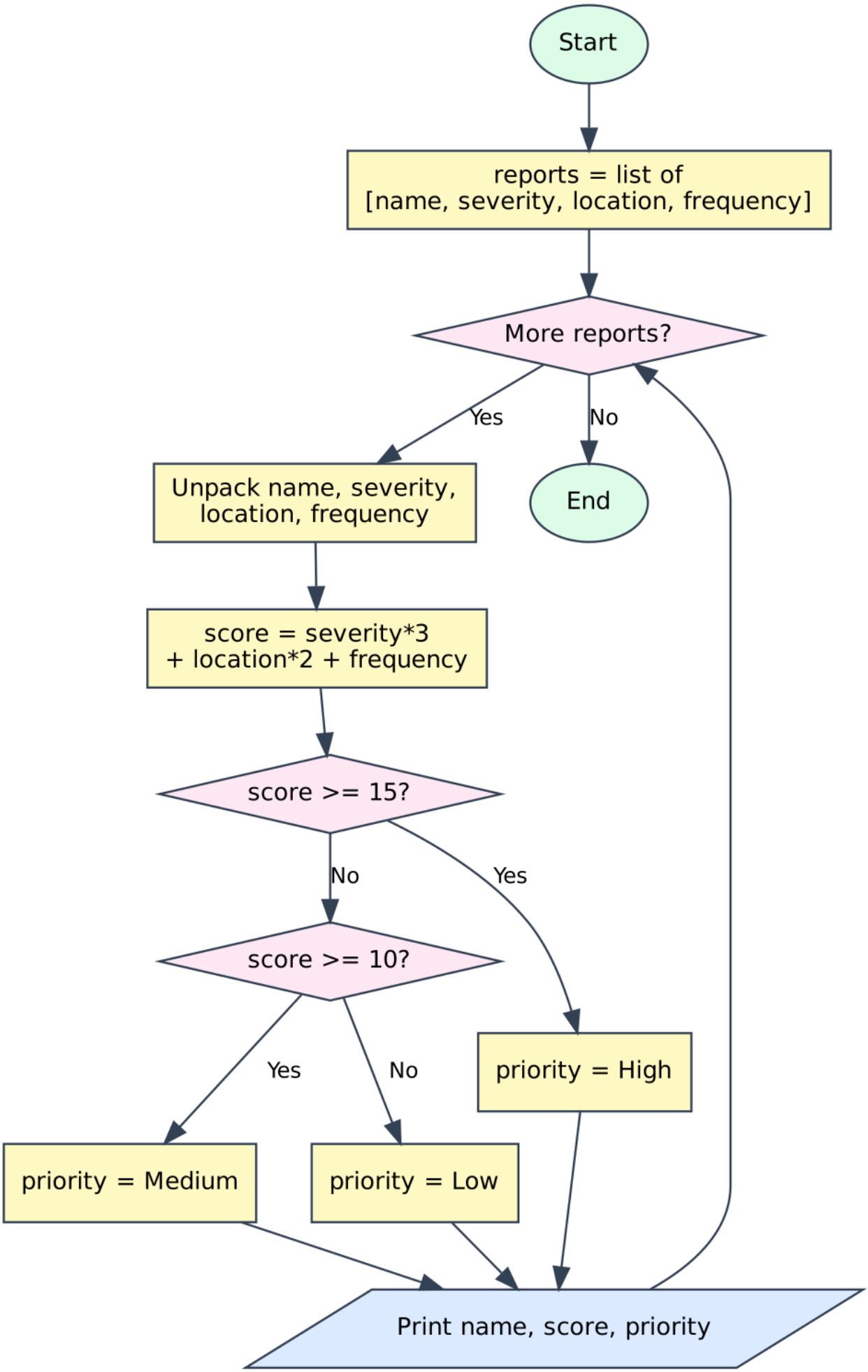
Task 1 - Library Resource Distribution

Output screenshot



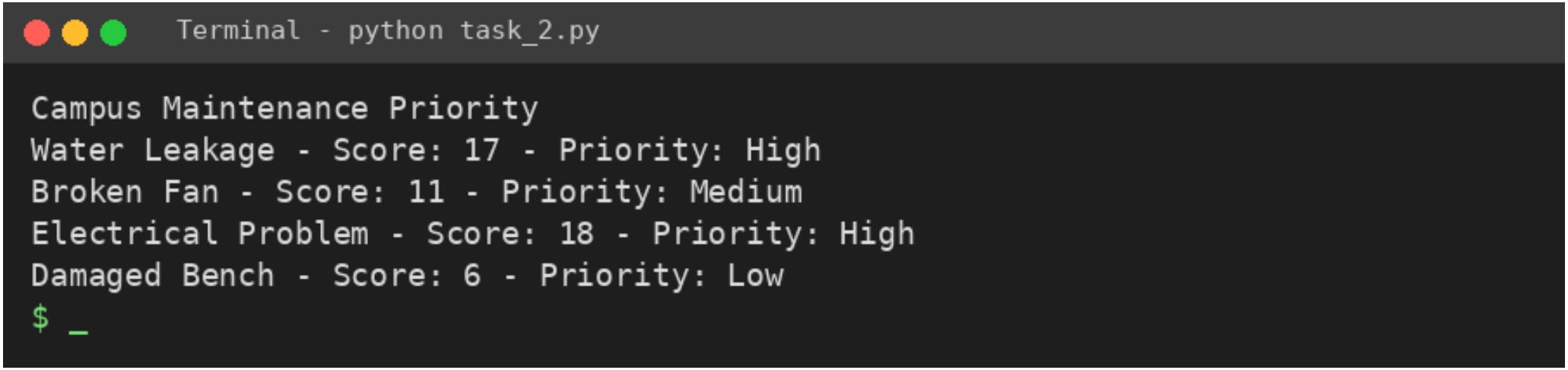
Task 2 - Campus Maintenance Priority

Flowchart



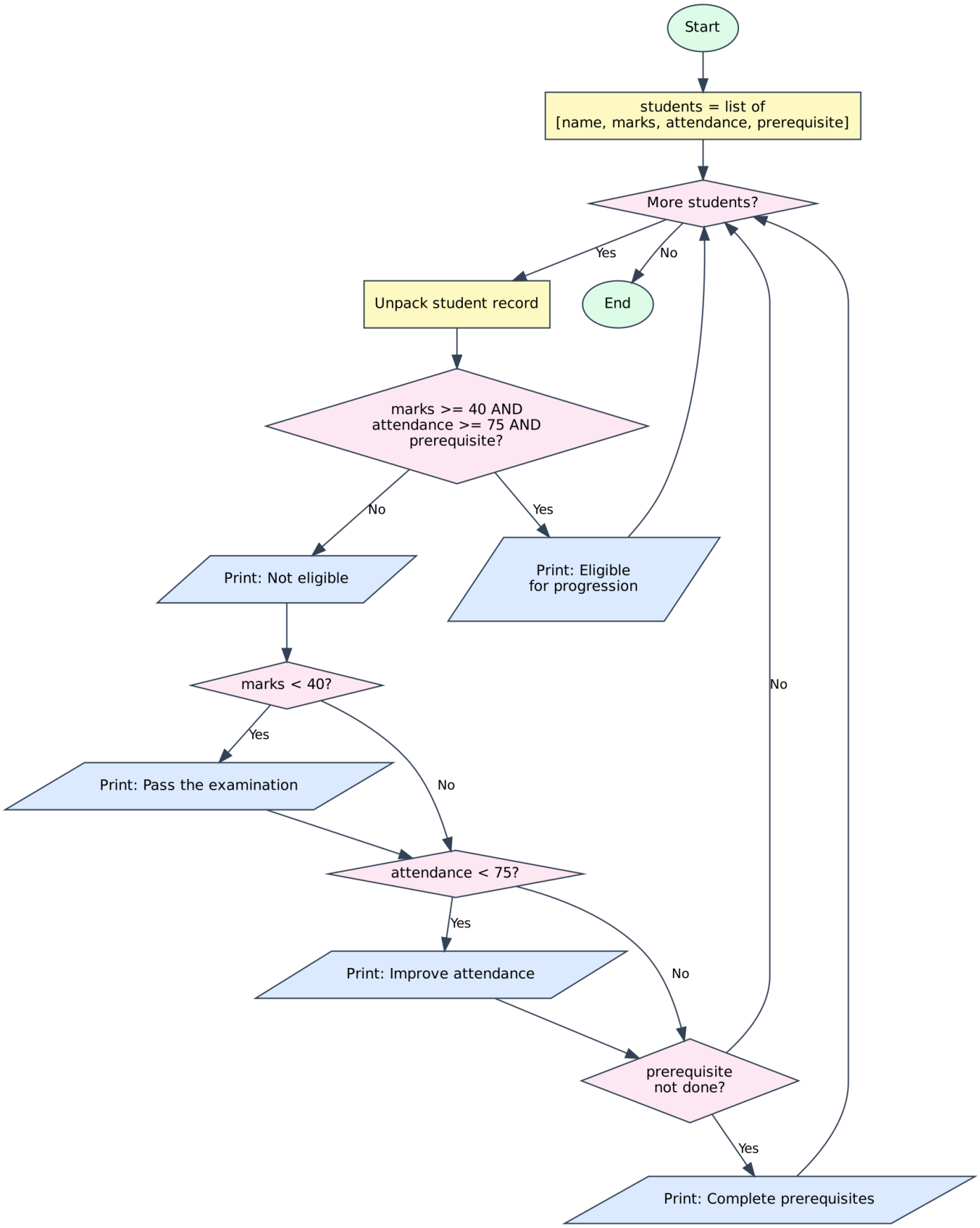
Task 2 - Campus Maintenance Priority

Output screenshot



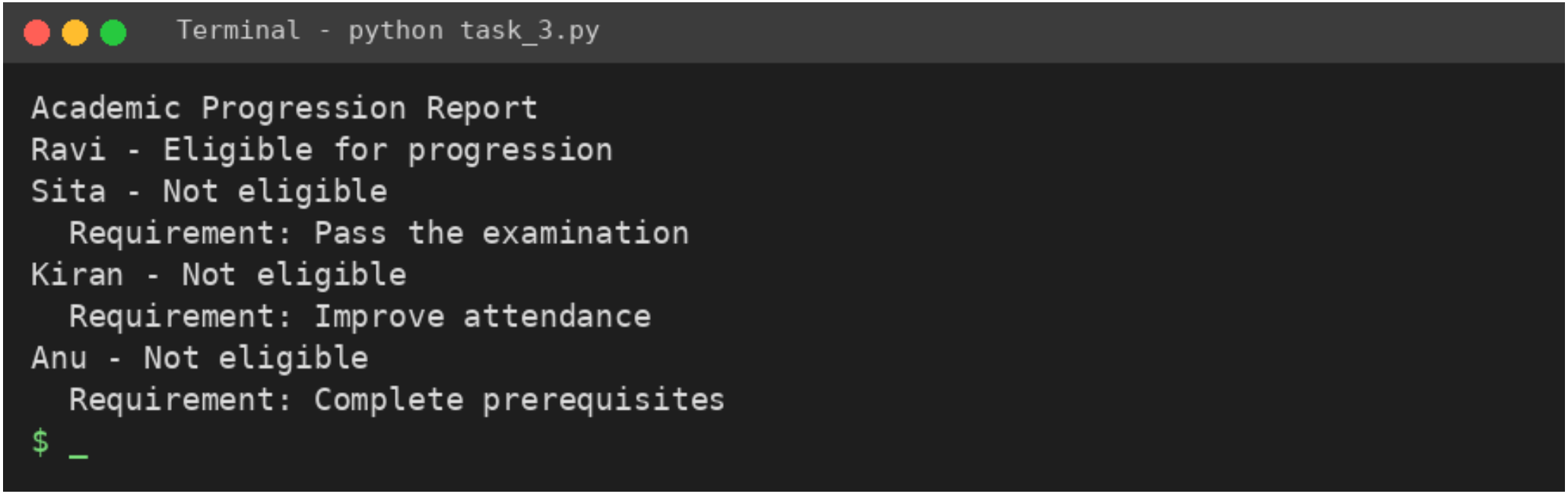
Task 3 - Academic Progression Report

Flowchart



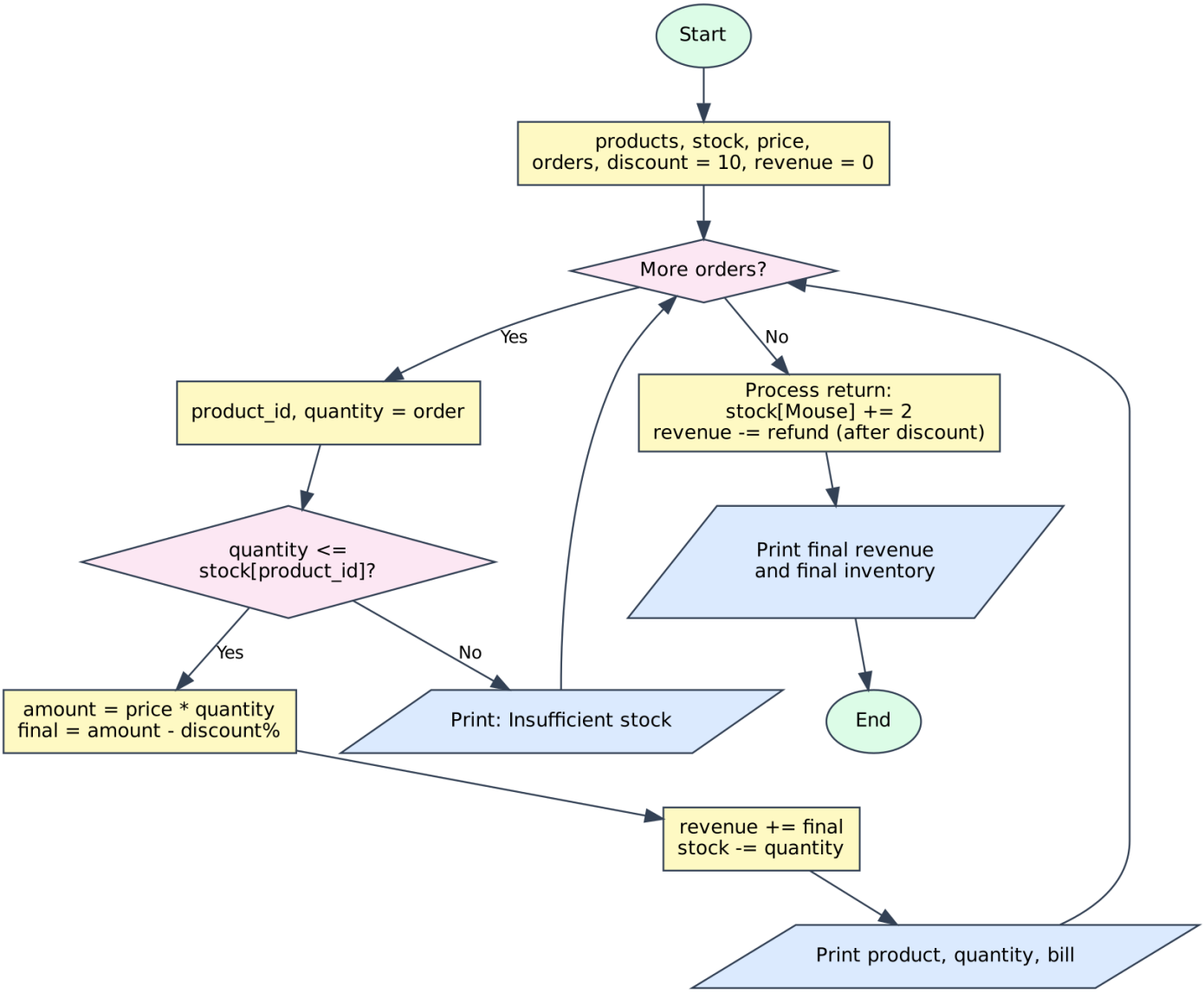
Task 3 - Academic Progression Report

Output screenshot



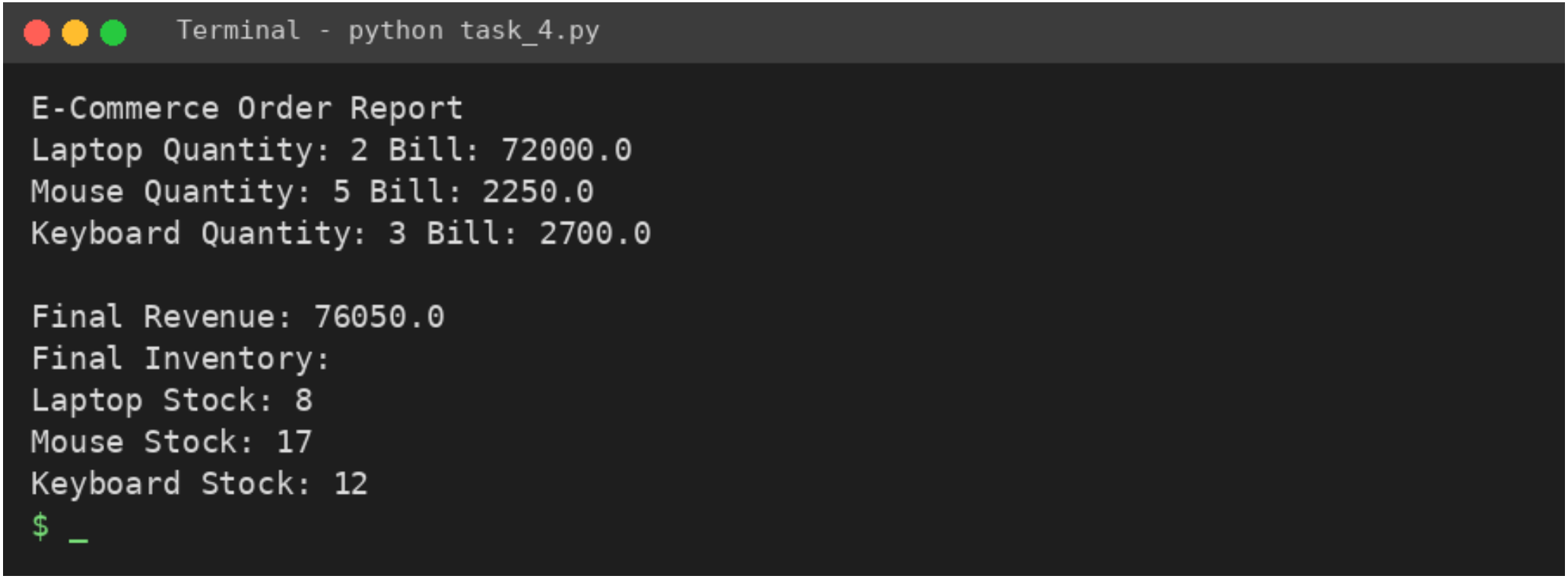
Task 4 - E-Commerce Order Report

Flowchart



Task 4 - E-Commerce Order Report

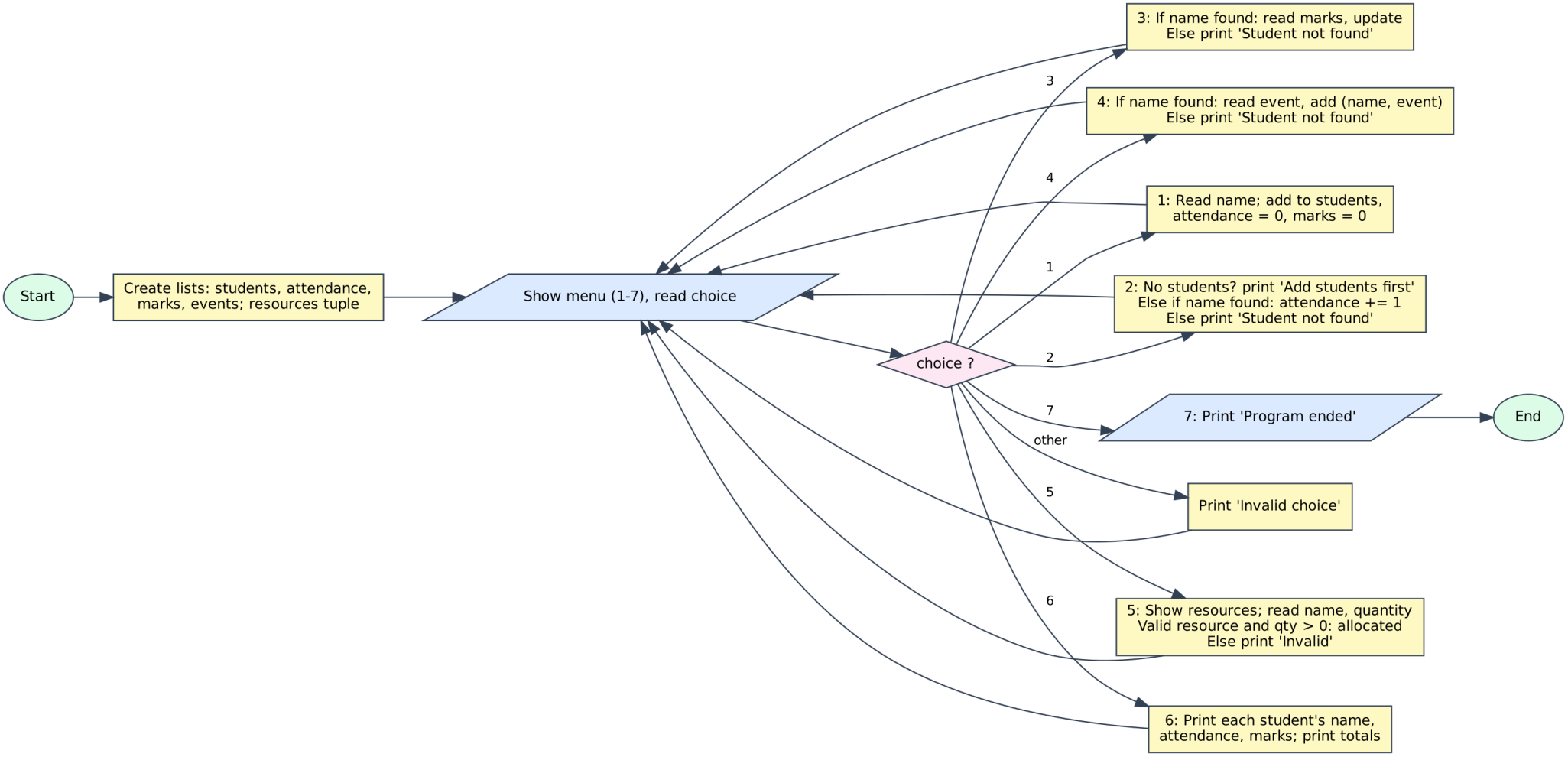
Output screenshot



Task 5 - Smart Campus Management (menu program)

Flowchart

Sample run: invalid choice, attendance before adding a student, add 2 students, attendance, marks, event, resource allocation, summary, exit.



Task 5 - Smart Campus Management (menu program)

Output screenshot (part 1 of 4)



Task 5 - Smart Campus Management (menu program)

Output screenshot (part 2 of 4)

```
3. Enter Marks
4. Register for Event
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 2
Enter student name: Ravi
Attendance recorded

--- SMART CAMPUS MANAGEMENT ---
1. Add Student
2. Record Attendance
3. Enter Marks
4. Register for Event
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 2
Enter student name: Ravi
Attendance recorded

--- SMART CAMPUS MANAGEMENT ---
1. Add Student
2. Record Attendance
3. Enter Marks
4. Register for Event
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 3
Enter student name: Ravi
Enter marks: 78
Marks updated

--- SMART CAMPUS MANAGEMENT ---
1. Add Student
2. Record Attendance
3. Enter Marks
4. Register for Event
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 3
Enter student name: Anu
Student not found

--- SMART CAMPUS MANAGEMENT ---
1. Add Student
2. Record Attendance
3. Enter Marks
4. Register for Event
```

Task 5 - Smart Campus Management (menu program)

Output screenshot (part 3 of 4)

```
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 50
Invalid choice. Try again.

--- SMART CAMPUS MANAGEMENT ---
1. Add Student
2. Record Attendance
3. Enter Marks
4. Register for Event
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 4
Enter student name: Ravi
Enter event name: Tech Fest
Event registration successful

--- SMART CAMPUS MANAGEMENT ---
1. Add Student
2. Record Attendance
3. Enter Marks
4. Register for Event
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 5
Available resources: ('Books', 'Projectors', 'Computers')
Enter resource name: Projectors
Enter quantity: 3
3 Projectors allocated successfully

--- SMART CAMPUS MANAGEMENT ---
1. Add Student
2. Record Attendance
3. Enter Marks
4. Register for Event
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 5
Available resources: ('Books', 'Projectors', 'Computers')
Enter resource name: Laptops
Enter quantity: 2
Invalid resource or quantity

--- SMART CAMPUS MANAGEMENT ---
1. Add Student
2. Record Attendance
3. Enter Marks
4. Register for Event
```

Task 5 - Smart Campus Management (menu program)

Output screenshot (part 4 of 4)

```
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 6

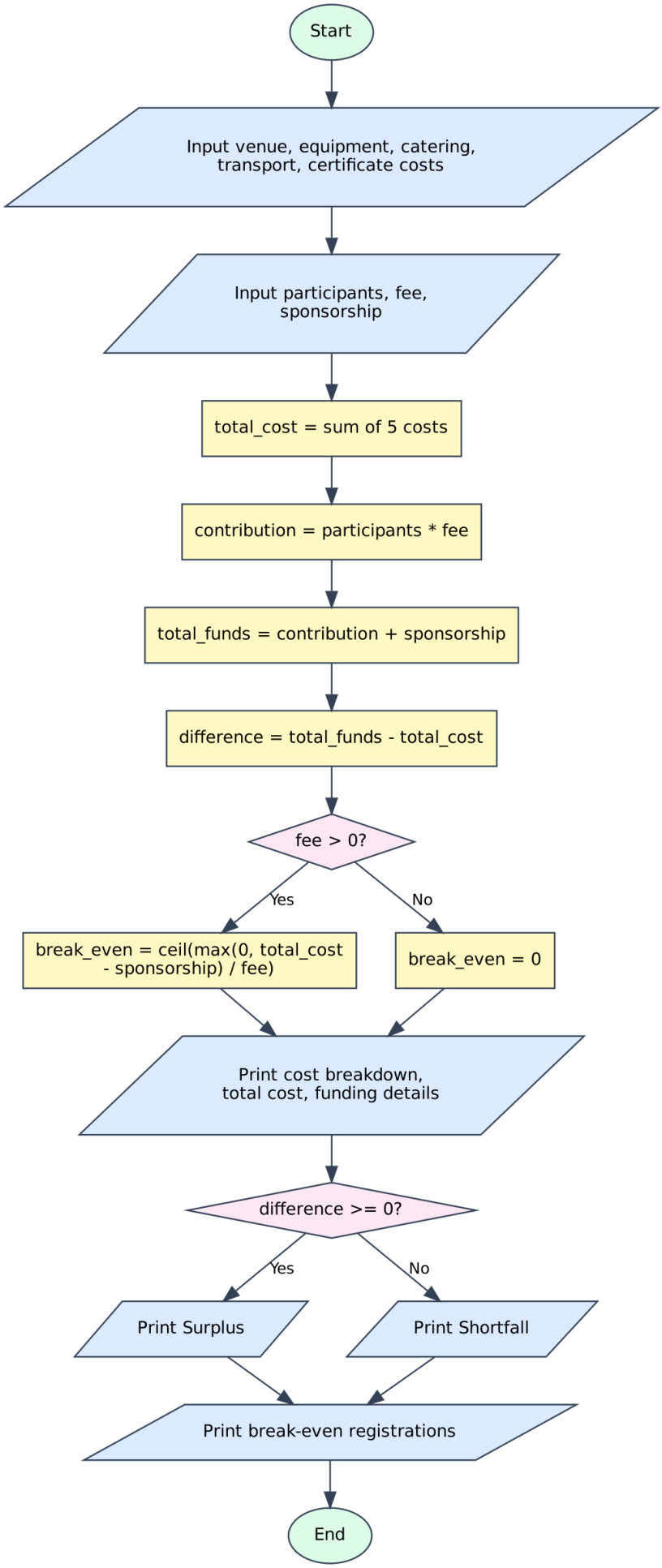
--- CAMPUS SUMMARY ---
Name: Ravi
Attendance: 2
Marks: 78.0
Name: Sita
Attendance: 0
Marks: 0
Total Students: 2
Event Registrations: 1
Resources: ('Books', 'Projectors', 'Computers')

--- SMART CAMPUS MANAGEMENT ---
1. Add Student
2. Record Attendance
3. Enter Marks
4. Register for Event
5. Allocate Resource
6. View Summary
7. Exit
Enter your choice: 7
Program ended
$ _
```

Task 6 - Campus Event Financial Planning

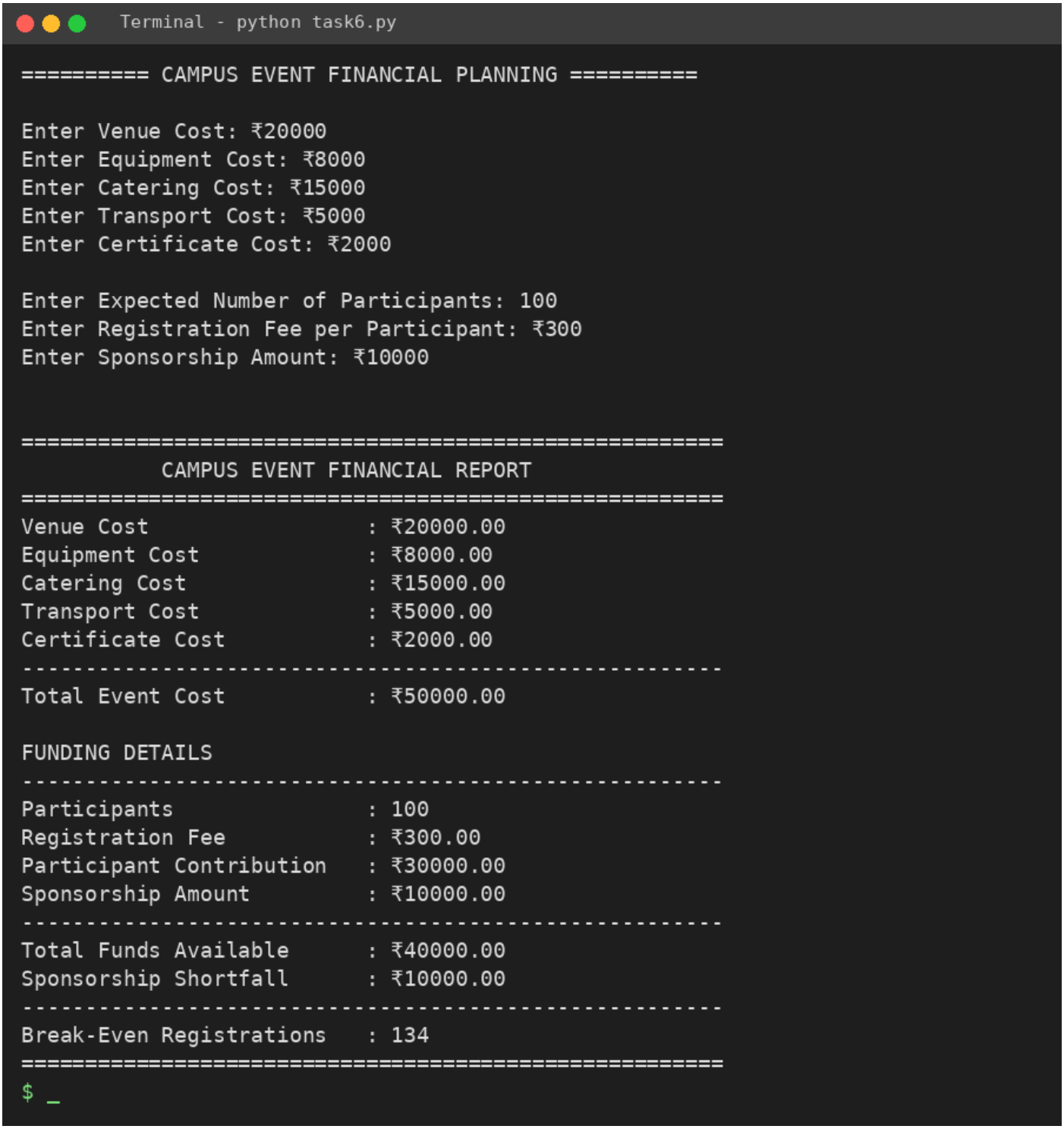
Flowchart

Note: the original file has '#import math' commented out, so it crashes with NameError. Output below is from a copy with 'import math' enable



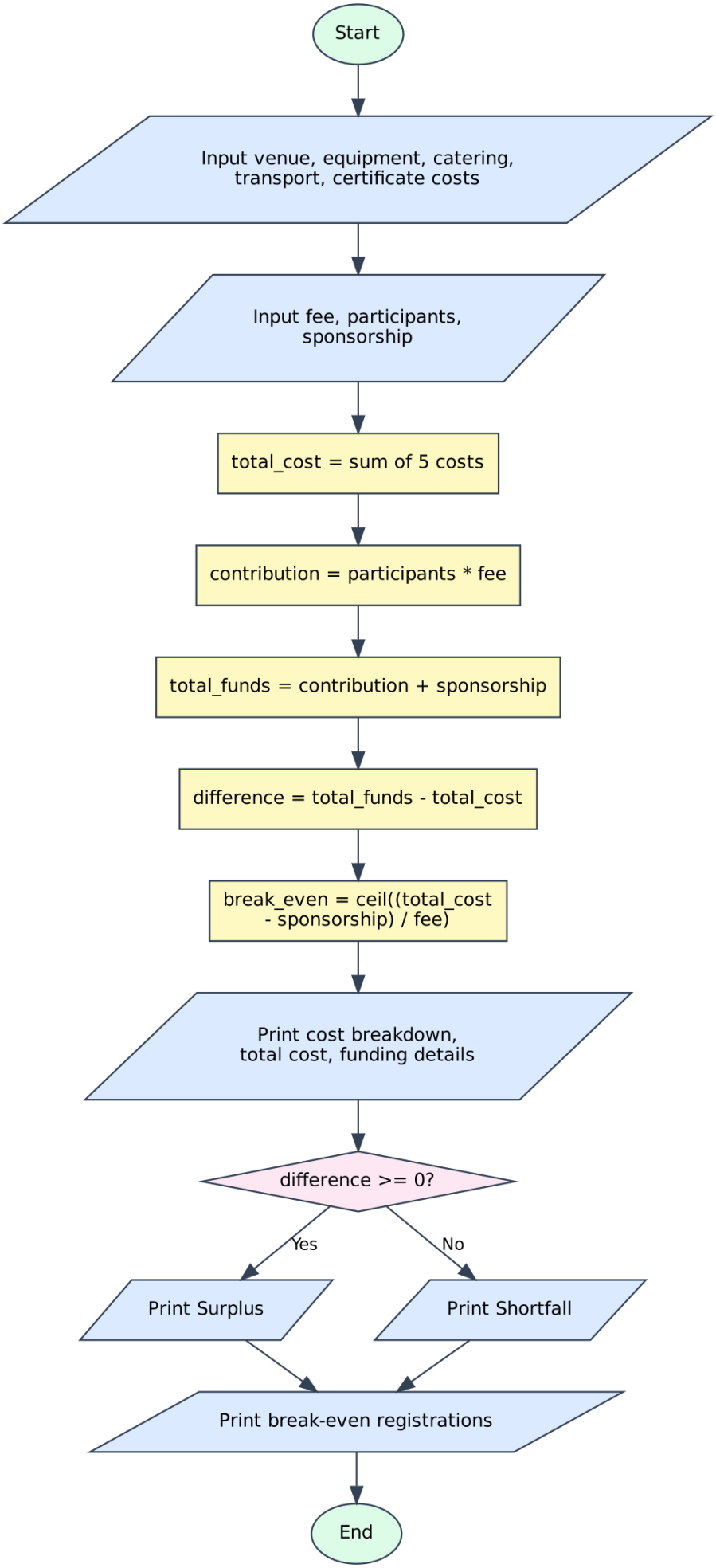
Task 6 - Campus Event Financial Planning

Output screenshot



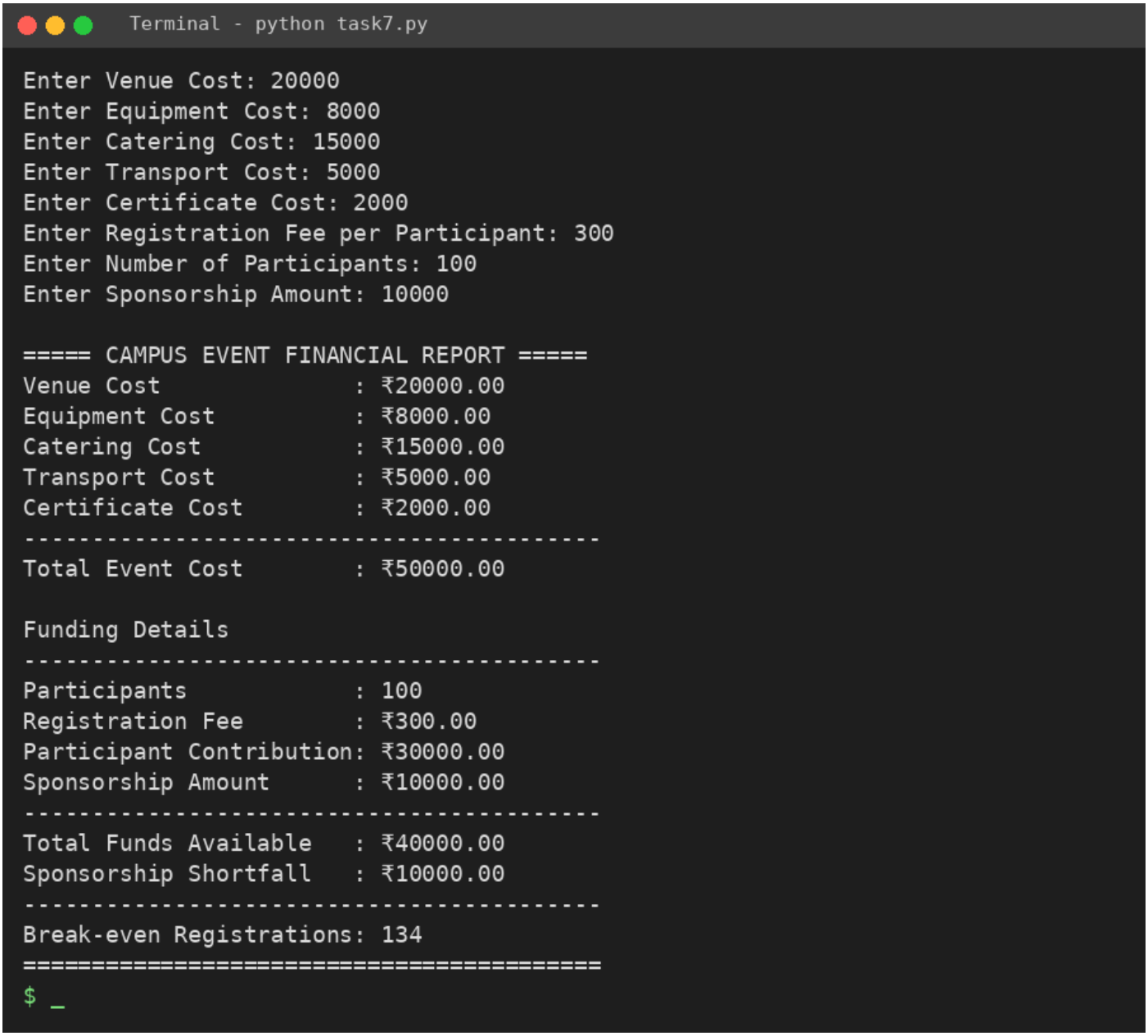
Task 7 - Campus Event Financial Report

Flowchart



Task 7 - Campus Event Financial Report

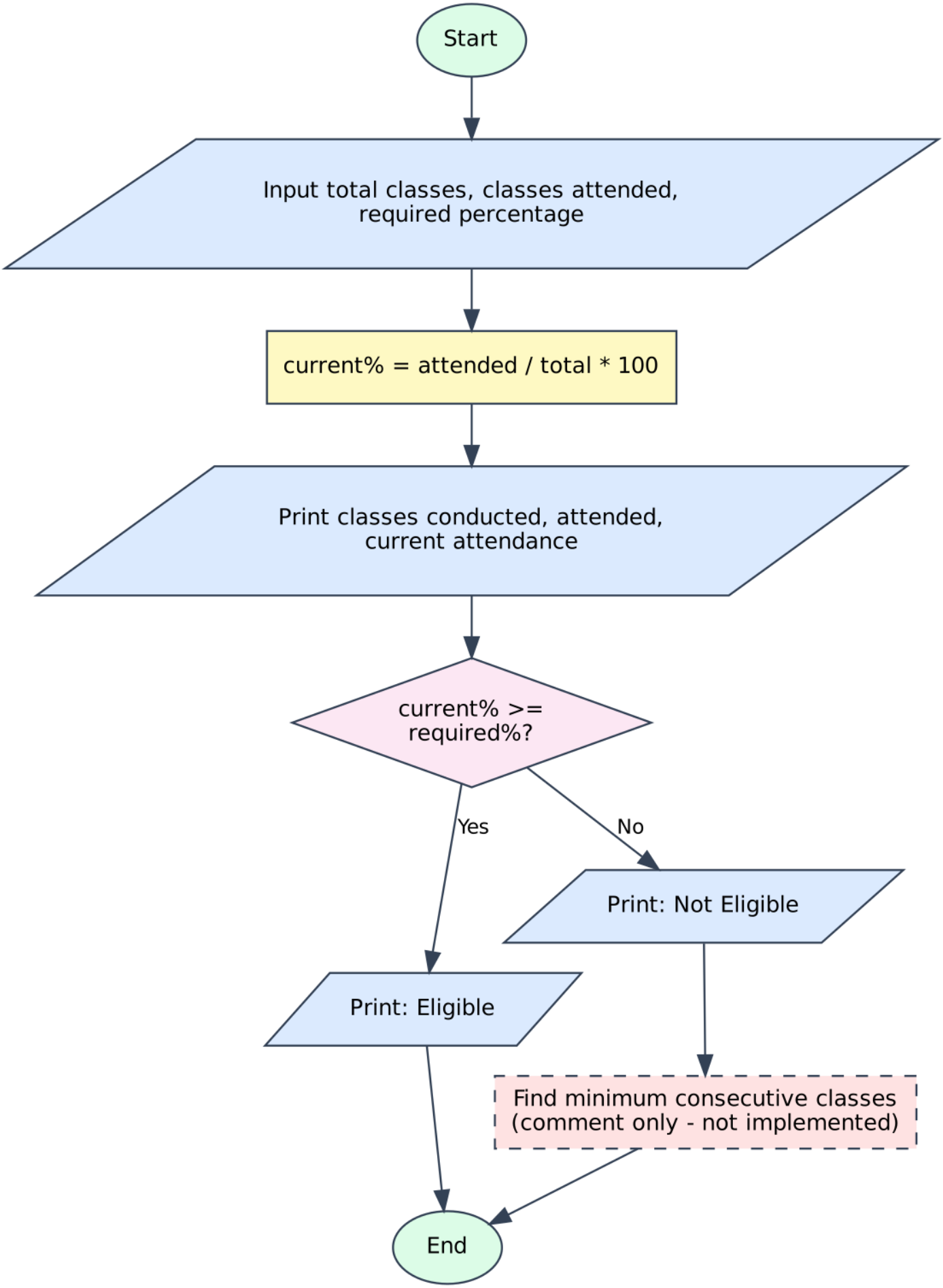
Output screenshot



Task 8 - Attendance Recovery Computation

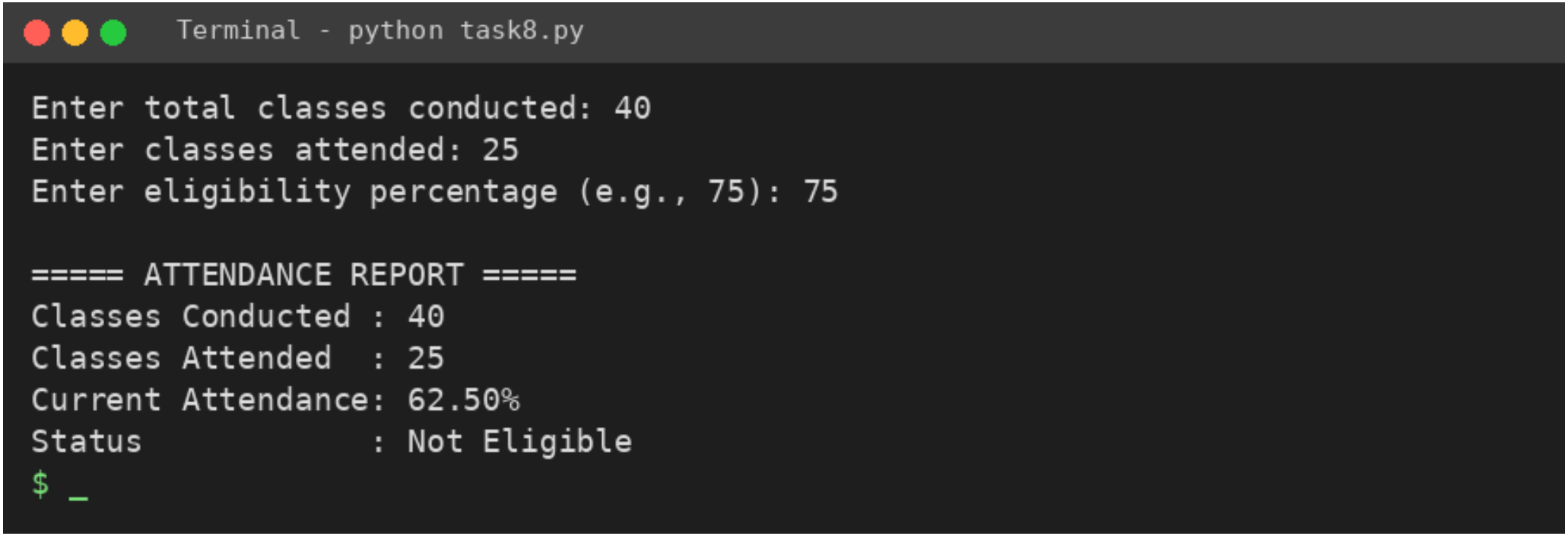
Flowchart

Note: the file is unfinished. It ends at the comment 'Find minimum consecutive classes', so only the report and eligibility check run.



Task 8 - Attendance Recovery Computation

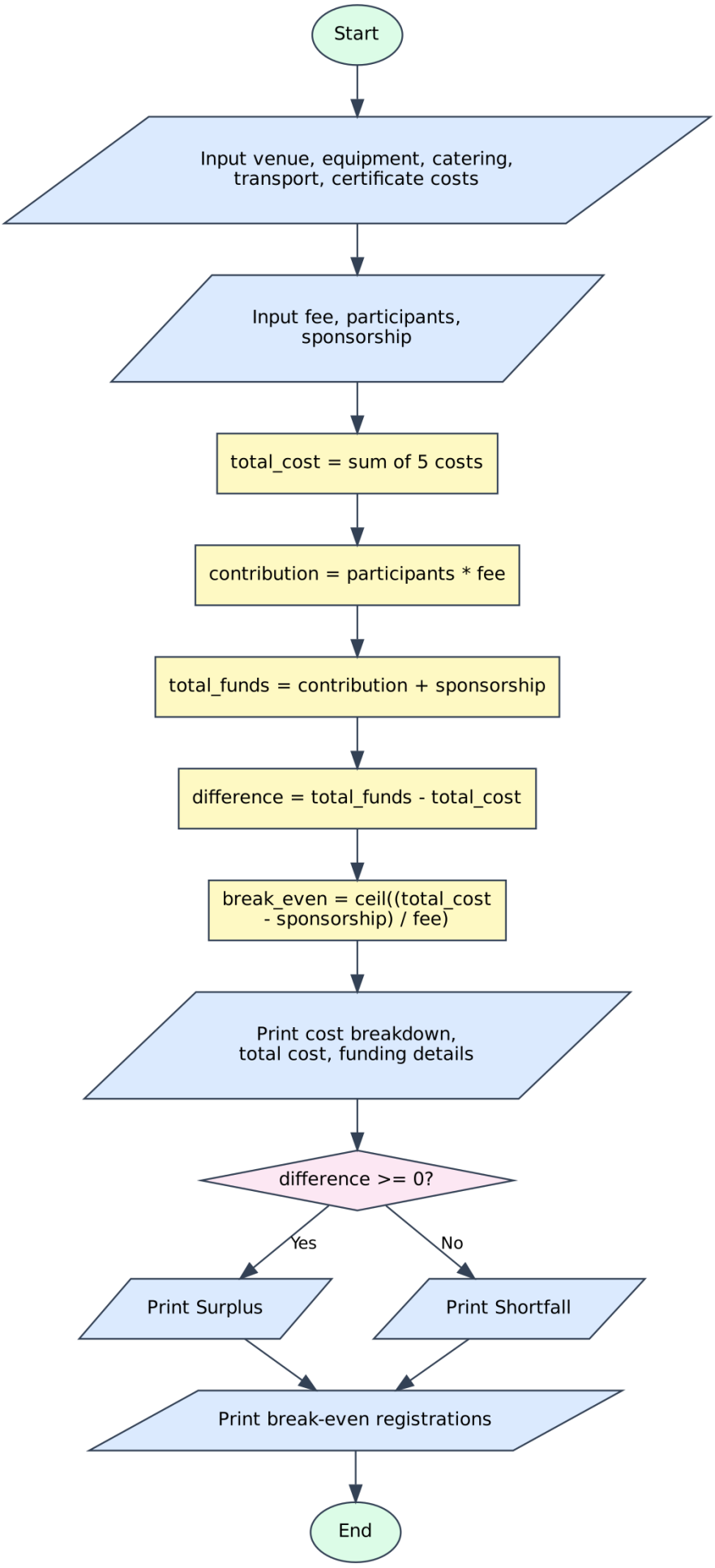
Output screenshot



Task 9 - Campus Event Financial Report

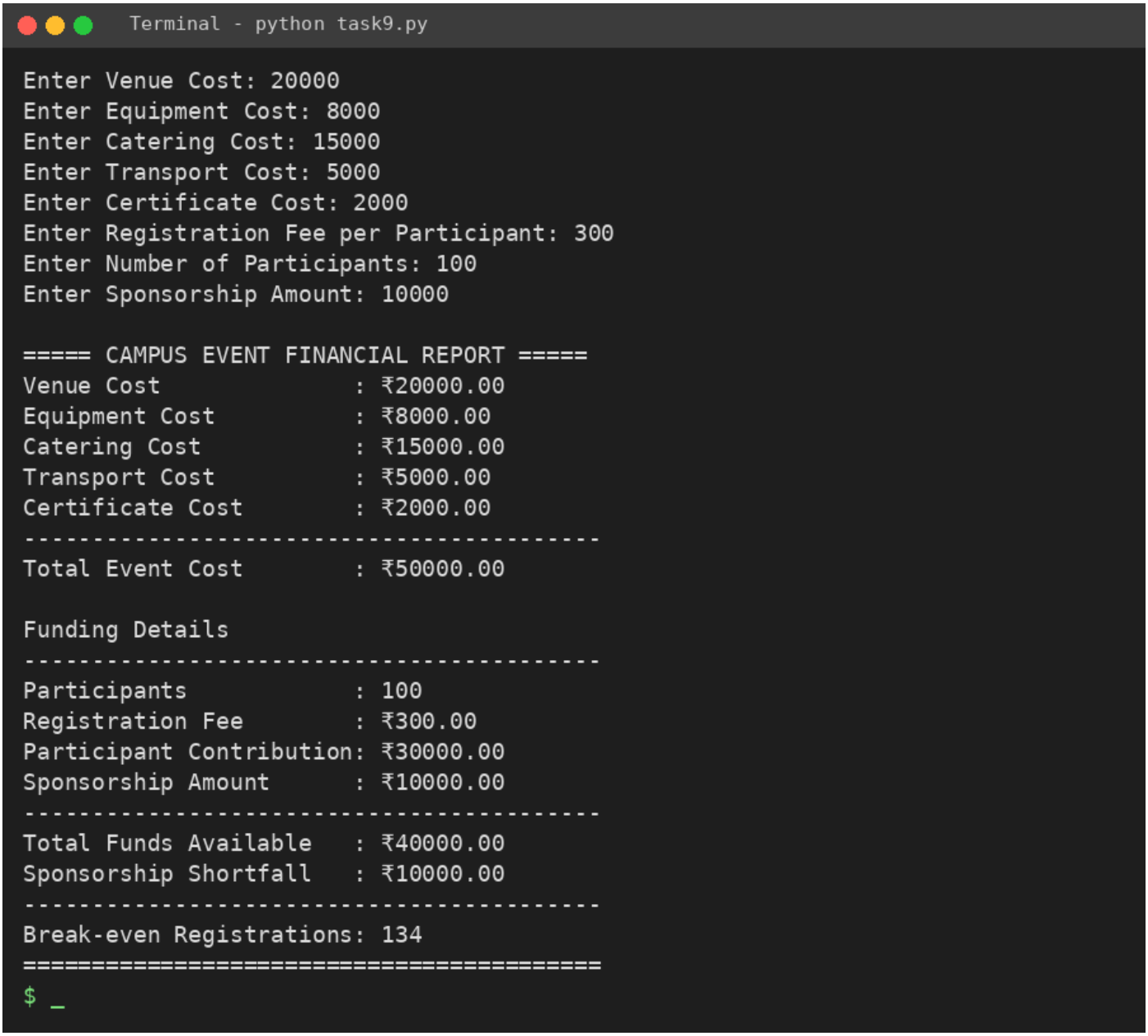
Flowchart

Note: identical code to Task 7.



Task 9 - Campus Event Financial Report

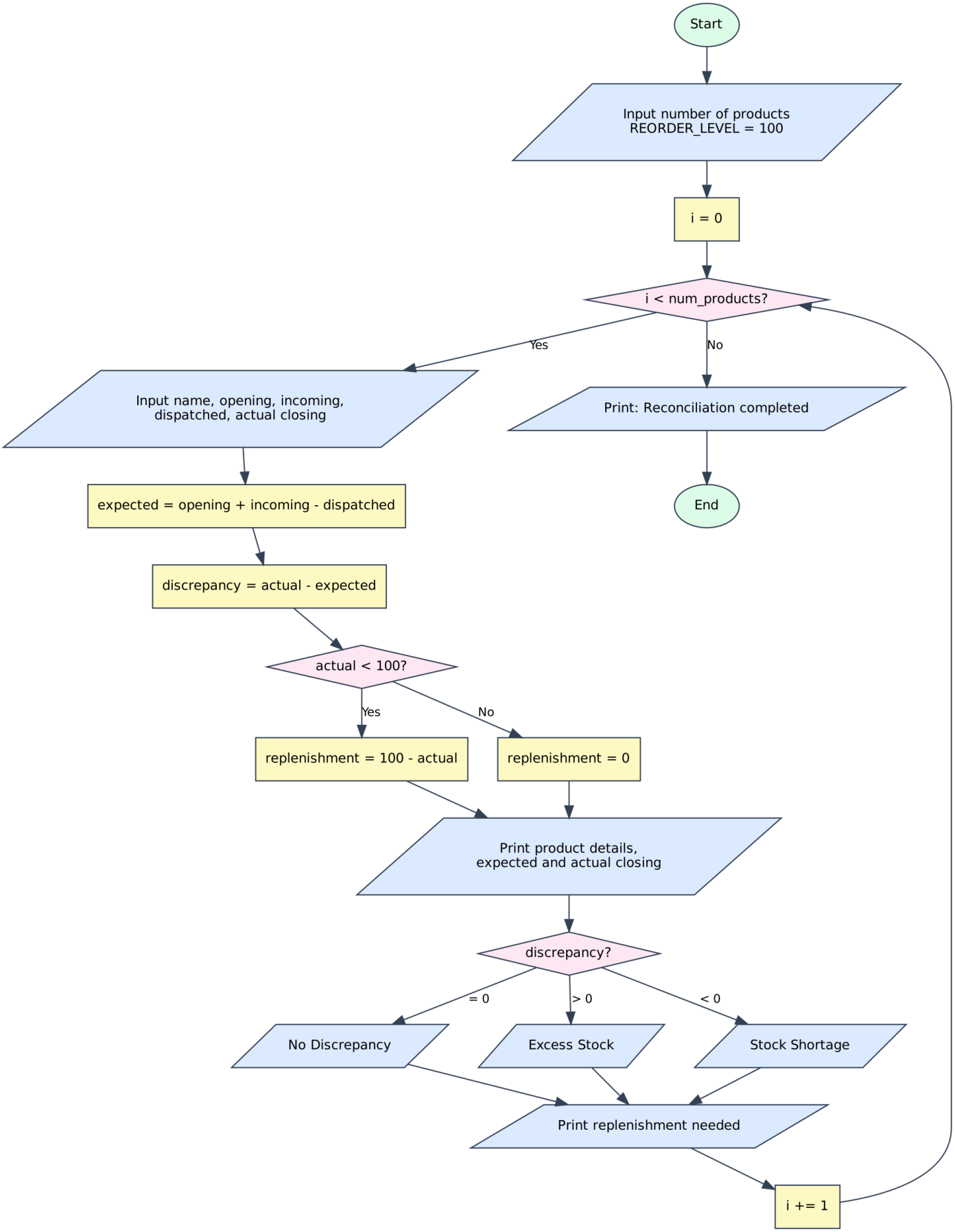
Output screenshot



Task 10 - Warehouse Inventory Reconciliation

Flowchart

Sample: Rice Bags (excess 5, no reorder) and Sugar (shortage 10, reorder 55).



Task 10 - Warehouse Inventory Reconciliation

Output screenshot

